Paper 24 — From Structural Diagnostics to Field-Theoretic Selection Functionals

A Maximum-Entropy and Large-Deviation Closure

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Abstract

The structural-selection programme introduces a functional $\mathcal{F}_{\text{struct}}$ to distinguish dynamically admissible configurations from configurations that are preferentially realised. In previous papers this functional was used phenomenologically: it organised numerical results and provided a global selection landscape, but its derivation from more primitive principles remained open.

This paper gives a field-theoretic route to $\mathcal{F}_{\text{struct}}$. The central claim is not that a final derivation from an ultimate microscopic theory has already been completed, but that the mathematical form of the structural functional follows from four basic requirements: dynamical admissibility, coarse-grained structural observability, maximum relative entropy, and large-deviation concentration. The Boltzmann-type selection weight

$$\mathbb{P}[X] \propto \exp[-\beta \mathcal{F}_{\text{struct}}[X]]$$

is not an independent postulate. It arises as the least biased probability measure over admissible configurations subject to fixed structural constraints.

Consequently, the structural functional must take the effective master form:

$$\mathcal{F}_{\text{struct}}[X] = -\log \frac{\mathrm{d}\mathbb{P}_{\text{sel}}}{\mathrm{d}\mu_0}[X] = \sum_a \lambda_a r_a[X],$$

up to an additive constant and a choice of primitive structural diagnostics r_a. In local field systems this becomes

$$\mathcal{F}_{\text{struct}}[X] = \int_M \mathrm{d}^d x \, \sqrt{|g|} \sum_a \lambda_a \rho_a(x; X).$$

Thus the structural functional is interpreted as a coarse-grained large-deviation rate functional for structural defects. We then give a canonical field-theoretic closure for local field theories: the primitive diagnostics are generated from equation residuals, symmetry-constraint residuals, coarse-grained activity, correlation loss, Hessian instability, and integrity defects of the base theory. This closes the construction at the level of effective local field theory; domain-specific work remains in the reference measure, coarse-graining scale, and calibration of the conjugate multipliers.

1 Motivation

Structural Selection Theory separates two questions:

What is dynamically admissible? and What is preferentially realised?

The first question is answered by a base dynamical law, for example an action S_0 and its equations of motion. The second question is answered by a selection functional $\mathcal{F}_{\text{struct}}$.

Previous work used the structural functional in the effective form

$$\mathbb{P}[X] \propto \exp[-\beta \mathcal{F}_{\text{struct}}[X]],$$

where X denotes a field configuration, history, background, or abstract state. This form was phenomenologically useful, but the origin of $\mathcal{F}_{\text{struct}}$ itself remained a central open problem.

The present paper addresses this problem at the effective-theory level. The goal is not to derive a unique microscopic formula for every domain. Rather, the goal is to show that the *form* of $\mathcal{F}_{\text{struct}}$ follows, under explicit structural constraints, from standard first principles once one accepts that selection acts on coarse-grained structural information.

2 Four First Principles

We assume four principles.

2.1 Principle 1: Dynamical Admissibility

Let $\mathcal{A}$ be the space of admissible configurations. Depending on the domain, this may be:

- the solution manifold of a field equation,
- the space of near-solutions below a residual tolerance,
- a gauge-equivalence class of string backgrounds,
- or the feasible set of an optimisation problem.

The base theory supplies a reference measure μ_0 on $\mathcal{A}$. In field theory this measure may be induced by a path-integral measure, Liouville measure, lattice ensemble, or a gauge-fixed coarse-grained measure [13, 14, 15]. The exact microscopic origin of μ_0 is domain-dependent, but the selection layer begins only after $\mathcal{A}$ and μ_0 have been specified.

For the minimal ϕ^4 lattice benchmark, a concrete choice is available: μ_0 may be taken as the regulated product measure over Euclidean lattice field values, restricted to the chosen boundary or topological sector. In a Hamiltonian lattice formulation this measure is extended to include canonical momenta. In the stochastic formulation, μ_0 is instead the stationary measure of the uncontrolled Langevin dynamics before structural reweighting is applied [10, 11]. In all cases, $\mathcal{F}_{\text{struct}}$ should be read as a selection density relative to this baseline ensemble, not as a replacement for the baseline dynamics.

2.2 Principle 2: Structural Observability

A coarse-grained observer or effective theory does not access every microscopic detail of $X \in \mathcal{A}$. It accesses a finite set of invariant structural diagnostics:

$$\mathbf{r}[X] = (r_1[X], \ldots, r_m[X]).$$

Each $r_a[X]$ is a nonnegative, dimensionless structural defect or cost. Typical examples include equation residuals, conservation defects, coherence loss, stability loss, boundary defects, anomaly residuals, or integrity defects.

The diagnostics must satisfy five admissibility conditions:

1. **Dimensionless boundedness:** each diagnostic can be normalised.
2. **Symmetry respect:** gauge, coordinate, and internal symmetries are not broken by the diagnostic.

3. **Orientation fidelity:** larger defect means worse structural performance.
4. **Non-redundancy:** no diagnostic should duplicate another one without adding information.
5. **Computability:** each diagnostic must be evaluable in concrete models.

2.3 Principle 3: Maximum Relative Entropy

Given only the expected structural diagnostics, the selected distribution over admissible configurations should be the least biased distribution relative to the reference measure μ_0. This is the maximum relative entropy principle [1, 2, 3, 4].

For a probability density $p[X]$ with respect to μ_0, define

$$S[p|\mu_0] = -\int_{\mathcal{A}} \mathrm{d}\mu_0(X)\, p[X] \log p[X].$$

We maximise this entropy subject to

$$\int_{\mathcal{A}} \mathrm{d}\mu_0(X)\, p[X] = 1, \qquad \int_{\mathcal{A}} \mathrm{d}\mu_0(X)\, p[X]\, r_a[X] = \bar{r}_a.$$

2.4 Principle 4: Large-Deviation Selection

For systems with many weakly correlated cells, modes, sectors, or degrees of freedom, probabilities of coarse-grained structural macrostates concentrate in the large-system limit:

$$\mathbb{P}_N(\mathbf{z}) \asymp \exp[-NI(\mathbf{z})].$$

The rate function $I(\mathbf{z})$ determines which macrostates are statistically dominant [7, 8, 9]. Structural selection is therefore identified with minimisation of a rate functional.

3 Maximum-Entropy Derivation

We now solve the variational problem. Introduce Lagrange multipliers α and λ_a and vary

$$\mathcal{L}[p] = -\int \mathrm{d}\mu_0\, p \log p - \alpha \left(\int \mathrm{d}\mu_0\, p - 1 \right) - \sum_a \lambda_a \left(\int \mathrm{d}\mu_0\, p\, r_a - \bar{r}_a \right).$$

Stationarity with respect to p gives

$$-(\log p + 1) - \alpha - \sum_a \lambda_a r_a[X] = 0.$$

Hence

$$p_*[X] = \frac{1}{Z(\lambda)} \exp\left[-\sum_a \lambda_a r_a[X] \right],$$

where

$$Z(\lambda) = \int_{\mathcal{A}} \mathrm{d}\mu_0(X)\, \exp\left[-\sum_a \lambda_a r_a[X] \right].$$

Under these assumptions, the effective selection measure must take the form

$$\mathrm{d}\mathbb{P}_{\mathrm{sel}}(X) = \frac{1}{Z(\lambda)} \exp[-\mathcal{F}_{\mathrm{struct}}[X]]\, \mathrm{d}\mu_0(X),$$

with

$$\boxed{\mathcal{F}_{\text{struct}}[X] = \sum_a \lambda_a r_a[X]} \tag{1}$$

up to an irrelevant additive constant.

Thus the structural functional is not introduced as an arbitrary new energy. It is the negative logarithm of the least biased selection weight relative to the admissible reference measure:

$$\boxed{\mathcal{F}_{\text{struct}}[X] = -\log \frac{\mathrm{d}\mathbb{P}_{\text{sel}}}{\mathrm{d}\mu_0}[X] \;+\; \text{const.}} \tag{2}$$

4 Local Field-Theoretic Form

If the structural diagnostics are local or quasi-local, each r_a can be written as

$$r_a[X] = \int_M \mathrm{d}^d x \, \sqrt{|g|} \, \rho_a(x; X),$$

where ρ_a is a scalar density built from the fields, their derivatives, and the background geometry.

Equation (1) then becomes

$$\boxed{\mathcal{F}_{\text{struct}}[X] = \int_M \mathrm{d}^d x \, \sqrt{|g|} \, U_{\text{struct}}(x; X)} \tag{3}$$

with

$$U_{\text{struct}}(x; X) = \sum_a \lambda_a \rho_a(x; X). \tag{4}$$

Nonlinear versions of U_{struct} arise when the constraints are not linear in the primitive diagnostics, or when diagnostics are first mapped through a closure operator before the maximum-entropy step. For example, if raw defects d_a are converted into bounded costs via

$$r_a[X] = -\log\left(\epsilon + \Gamma_a[X]\right), \qquad \Gamma_a[X] \in [0, 1],$$

then Eq. (1) becomes

$$\mathcal{F}_{\text{struct}}[X] = -\sum_a \lambda_a \log(\epsilon + \Gamma_a[X]).$$

This form rewards configurations whose structural support scores Γ_a are close to one and penalises configurations with large structural defects.

5 Field-Theoretic Closure of the Primitive Diagnostics

We now close the remaining gap at the level of local field theory. The primitive diagnostics are not arbitrary scores. Given a base action, a symmetry class, and a coarse-graining prescription, they are generated by the minimal defects that obstruct a configuration from being dynamically consistent, symmetric, stable, coherent, and well formed.

5.1 Field-Theoretic Data

Let the field-theoretic input be the quadruple

$$\mathfrak{D}_{\text{micro}} = \left(S_0[\mathbf{\Phi}],\, G,\, \Pi_\ell,\, \mu_0\right),$$

where:

- $S_0[\mathbf{\Phi}]$ is the base action for fields $\mathbf{\Phi} = (\Phi_1, \ldots, \Phi_n)$,
- G is the gauge, coordinate, or internal symmetry group,
- Π_ℓ is a coarse-graining map at scale ℓ,
- μ_0 is the reference measure on admissible or near-admissible configurations.

The fluctuation operator is

$$\mathcal{K}_{ij}(x, y) = \frac{\delta^2 S_0}{\delta\Phi_i(x)\,\delta\Phi_j(y)}.$$

The Euler–Lagrange residuals are

$$E_i(x; \mathbf{\Phi}) = \frac{1}{\sqrt{|g|}} \frac{\delta S_0}{\delta\Phi_i(x)}.$$

All primitive diagnostics below are built from these objects and then normalised by intrinsic scales Λ_a.

5.2 Six Canonical Primitive Defects

1. Equation residual defect. The first obstruction is failure to satisfy the field equations of motion:

$$d_H[\mathbf{\Phi}] = \frac{1}{\Lambda_H} \int_M \mathrm{d}^d x \sqrt{|g|} \sum_i |\Pi_\ell E_i(x; \mathbf{\Phi})|^2 .$$

This is the field-theoretic origin of the harmony or equation-consistency sector.

2. Noether/Ward constraint defect. Symmetries imply local identities: conservation laws, Bianchi identities, Gauss constraints, gauge constraints, anomaly-cancellation conditions, or tadpole constraints [12]. Let $\mathcal{W}_\alpha[\mathbf{\Phi}]$ denote the residual of the α-th identity. Then

$$d_B[\mathbf{\Phi}] = \frac{1}{\Lambda_B} \int_M \mathrm{d}^d x \sqrt{|g|} \sum_\alpha |\Pi_\ell \mathcal{W}_\alpha(x; \mathbf{\Phi})|^2 .$$

For an ordinary relativistic field theory this includes $\mathcal{W}_\nu = \nabla_\mu T^\mu{}_\nu$; for gauge theories it includes the corresponding Ward or Gauss-law residuals.

3. Coarse-grained activity defect. The activity sector measures unresolved dynamical production at the coarse-grained level. Let $P_\ell(\mathbf{\Phi}, t)$ be the field-value distribution after applying Π_ℓ, and define

$$S_{\mathrm{cg}}[P_\ell] = -\int \mathrm{d}z\, P_\ell(z) \log P_\ell(z).$$

The positive activity rate is

$$A_\ell[\mathbf{\Phi}] = \max\left(0, \frac{\mathrm{d}}{\mathrm{d}t} S_{\mathrm{cg}}[P_\ell]\right).$$

As a defect relative to a controlled activity scale A_*, one may use

$$d_S[\mathbf{\Phi}] = \frac{1}{\Lambda_S} \left(A_\ell[\mathbf{\Phi}] - A_*\right)^2 .$$

For passive field benchmarks $A_* = 0$ gives a direct penalty for unresolved coarse-grained production. For active or creative regimes A_* may be nonzero, allowing controlled activity without treating all dynamics as bad.

In high-dimensional systems, estimating the full distribution P_ℓ can be expensive. The formal definition therefore admits practical proxies, such as local histogram updates, mode-energy transfer rates, positive changes of a coarse-grained entropy estimator, or local production residuals. Such proxies are admissible when they preserve the same orientation: larger unresolved activity corresponds to larger d_S [11].

4. Correlation-loss defect. Let $I_{\ell,ij}$ be the mutual information between neighbouring coarse-grained cells or coupled sectors. A simple field-theoretic correlation support is

$$I_\ell[\mathbf{\Phi}] = \sum_{\langle i,j \rangle} I_{\ell,ij}.$$

Using a reference coherence level I_*, the defect is

$$d_V[\mathbf{\Phi}] = \frac{1}{\Lambda_V} \left(\max(0, I_* - I_\ell[\mathbf{\Phi}])\right)^2 .$$

This measures loss of structural connectivity rather than coupling strength itself.

Full mutual-information estimation may also be costly outside small lattice models. In practice, I_ℓ can be approximated by nearest-neighbour mutual information, covariance-based Gaussian information estimates, correlation lengths, or graph-coupling summaries, provided the approximation is monotone with respect to the loss of structural coherence [5, 6].

5. Stability defect. The second variation determines whether a configuration is stable against small perturbations. Let $\lambda_k(\mathcal{K})$ be the spectrum of the fluctuation operator after gauge zero-modes have been removed. The field-theoretic stability defect is

$$d_R[\mathbf{\Phi}] = \frac{1}{\Lambda_R} \sum_k \left[\max(0, -\lambda_k(\mathcal{K}))\right]^2 .$$

This is a one-sided penalty: stable positive modes do not produce extra cost, while tachyonic or negative directions are penalised [16, 17].

6. Integrity and boundary defect. Finally, a configuration may satisfy local equations while failing global integrity conditions: boundary conditions, amplitude bounds, topological sector constraints, regularity conditions, or compactification consistency. Let $\mathcal{Q}_\beta[\mathbf{\Phi}]$ denote such residuals. Then

$$d_\Sigma[\mathbf{\Phi}] = \frac{1}{\Lambda_\Sigma} \sum_\beta \left|\mathcal{Q}_\beta[\mathbf{\Phi}]\right|^2 .$$

5.3 From Defects to Structural Support

The raw defects have a common orientation: larger means worse. Convert them into bounded structural support scores through

$$\Gamma_a[\mathbf{\Phi}] = \frac{1}{1 + d_a[\mathbf{\Phi}]}, \qquad a \in \{H, B, S, V, R, \Sigma\}.$$

The field-theoretic structural functional becomes

$$\boxed{\mathcal{F}_{\text{struct}}[\mathbf{\Phi}] = - \sum_{a \in \{H,B,S,V,R,\Sigma\}} \lambda_a \log(\epsilon + \Gamma_a[\mathbf{\Phi}])} \tag{5}$$

This logarithmic form is the natural one when each support score Γ_a is interpreted as a sector-wise probability of structural admissibility. In that case, independent sector supports multiply,

$$P_{\text{support}}[\mathbf{\Phi}] \propto \prod_a (\epsilon + \Gamma_a[\mathbf{\Phi}])^{\lambda_a},$$

and $\mathcal{F}_{\text{struct}}$ is exactly the negative log-support. As a small-defect or deliberately chosen direct-cost representation, one may also use

$$\mathcal{F}_{\text{struct}}[\mathbf{\Phi}] = \sum_a \lambda_a d_a[\mathbf{\Phi}].$$

The linear form is therefore not a general identity with the logarithmic form; it is a modelling choice or small-defect approximation when the raw defects themselves are taken as the primitive costs.

5.4 How the Multipliers Are Fixed

The multipliers are not arbitrary decoration. In the maximum-entropy construction they are conjugate to the structural constraints:

$$\bar{r}_a = -\frac{\partial}{\partial \lambda_a} \log Z(\lambda).$$

Thus λ_a can be fixed in three principled ways:

- by imposing target expectation values $\bar{r}_a$,
- by fitting the selected ensemble through moment matching,
- or by deriving them from a deeper bath, environment, or coarse-graining flow.

In all cases, changing λ_a changes the structural selection pressure conjugate to defect r_a.

5.5 Minimality Statement

For local field theories, the six sectors above define a minimal closure, not a unique or final list of all possible diagnostics. The minimality is operational: removing d_H ignores the equations of motion; removing d_B ignores symmetry constraints; removing d_S ignores unresolved activity; removing d_V ignores correlation structure; removing d_R ignores linear stability; and removing d_Σ ignores global integrity. Additional diagnostics may be useful in specialised models, but they should either refine one of these six sectors or demonstrate independent predictive content.

6 Large-Deviation Interpretation

The maximum-entropy derivation gives the canonical ensemble over configurations. The large-deviation principle explains why minimisation of $\mathcal{F}_{\text{struct}}$ becomes a selection rule.

Let $\mathbf{z} = \mathbf{r}[X]$ denote a structural macrostate. The induced probability of $\mathbf{z}$ is

$$\mathbb{P}(\mathbf{z}) = \int_{\mathcal{A}} \mathrm{d}\mathbb{P}_{\text{sel}}(X)\, \delta(\mathbf{r}[X] - \mathbf{z}).$$

For many effective degrees of freedom,

$$\mathbb{P}_N(\mathbf{z}) \asymp \exp[-NI(\mathbf{z})].$$

The rate function can be written as a Legendre-Fenchel transform of the log-partition function:

$$I(\mathbf{z}) = \sup_{\theta} \left[\theta \cdot \mathbf{z} - \log Z(\theta)\right].$$

The selected macrostate satisfies

$$\mathbf{z}_{\text{sel}} = \arg\min_{\mathbf{z}} I(\mathbf{z}).$$

At the configuration level, this becomes the familiar structural-selection rule:

$$X_{\text{sel}} = \arg\min_{X \in \mathcal{A}} \mathcal{F}_{\text{struct}}[X].$$

Thus $\mathcal{F}_{\text{struct}}$ is the configuration-level representative of a large-deviation rate functional.

7 Relation to the CCI

The Critical Coherence Index is a reduced instability observable, while $\mathcal{F}_{\text{struct}}$ is the primary global selection functional. The present derivation clarifies their relation without making the CCI part of the fundamental definition of $\mathcal{F}_{\text{struct}}$.

The CCI has the form

$$\mathcal{C}_{\text{CCI}} = \frac{\text{destabilising drive}}{\text{coherence-preserving support}}.$$

It is naturally ratio-like and diagnostic. By contrast, $\mathcal{F}_{\text{struct}}$ is additive and rate-like. A natural additive CCI cost is therefore

$$r_{\mathcal{C}_{\text{CCI}}}[X] = \log(1 + \mathcal{C}_{\text{CCI}}[X]).$$

If the CCI is optionally included as one of the structural constraints, the maximum-entropy derivation gives

$$\mathcal{F}_{\text{struct}}[X] = \lambda_{\mathcal{C}_{\text{CCI}}} \log(1 + \mathcal{C}_{\text{CCI}}[X]) + \sum_b \lambda_b r_b[X].$$

This explains why the CCI may correlate strongly with $\mathcal{F}_{\text{struct}}$ without being identical to it: the CCI can be used as one reduced diagnostic sector of the structural rate functional, while $\mathcal{F}_{\text{struct}}$ remains the more primary object containing balance, integrity, stability, and global-order information.

8 Scalar-Field Reduction

For a single scalar field ϕ with base action $S_0[\phi]$, the field-theoretic closure reduces to explicit scalar expressions. The equation residual defect is

$$d_H[\phi] = \frac{1}{\Lambda_H} \int \mathrm{d}^d x \left(\frac{\delta S_0}{\delta \phi}\right)^2,$$

and the conservation defect is

$$d_B[\phi] = \frac{1}{\Lambda_B} \int \mathrm{d}^d x \, (\nabla_\mu T^{\mu\nu})(\nabla_\rho T^\rho{}_\nu),$$

while the stability defect is determined by the fluctuation operator

$$\mathcal{K}(x, y) = \frac{\delta^2 S_0}{\delta\phi(x)\delta\phi(y)}.$$

$$d_R[\phi] = \frac{1}{\Lambda_R} \max(0, -\lambda_{\min}[\delta^2 S_0])^2,$$

with the obvious replacement by a sum over negative eigenvalues when several unstable directions are present. The remaining sectors are obtained from the same field-theoretic recipe:

$$d_S[\phi] = \frac{1}{\Lambda_S} \left(A_\ell[\phi] - A_*\right)^2, \qquad d_V[\phi] = \frac{1}{\Lambda_V} \left(\max(0, I_* - I_\ell[\phi])\right)^2,$$

$$d_\Sigma[\phi] = \frac{1}{\Lambda_\Sigma} \sum_\beta |\mathcal{Q}_\beta[\phi]|^2.$$

Here A_ℓ is the coarse-grained activity rate, I_ℓ is the coarse-grained correlation support, and $\mathcal{Q}_\beta$ are boundary, topological, amplitude, or regularity residuals.

Using bounded support scores

$$\Gamma_a[\phi] = \frac{1}{1 + d_a[\phi]},$$

one obtains

$$\mathcal{F}_{\text{struct}}[\phi] = -\sum_a \lambda_a \log(\epsilon + \Gamma_a[\phi]).$$

This recovers the qualitative structure used in the ϕ^4 benchmark: true vacua, stable kinks, and coherent low-defect configurations receive lower structural cost than off-shell, unstable, or distorted configurations.

9 What Has Been Closed

The effective derivation closes four points that were previously postulated:

- the exponential selection weight follows from maximum relative entropy;
- the structural functional is the negative log-density of the selected measure relative to the admissible reference measure;
- minimisation of $\mathcal{F}_{\text{struct}}$ follows from large-deviation concentration;
- in local field theories, the primitive sectors are generated from the action, symmetries, coarse-graining map, and fluctuation operator.

The Lagrange multipliers λ_a acquire a clear interpretation: they are selection pressures conjugate to structural defects. Their values are not universal constants. They depend on the ensemble, coarse-graining scale, and domain-specific constraints.

10 What Remains Open

This derivation should be read as an effective field-theoretic route, not as the final completion from an ultimate microscopic theory. Four problems remain:

1. The reference measure μ_0 must be specified in each domain.
2. The six canonical primitive diagnostics must be benchmarked and, where necessary, refined for specialised domains.
3. The multipliers λ_a must be inferred, fixed by constraints, or derived from deeper dynamics.
4. In quantum gravity or string theory, gauge redundancy, moduli measures, and background dependence require additional care.

The important change is that the open problem has been narrowed. The structural functional is no longer an arbitrary phenomenological object. Its general form follows from entropy maximisation and rate-function selection, and its local field-theoretic primitive diagnostics follow from the local consistency conditions of the base action. What remains is the calibration of the reference measure, the multipliers, and domain-specific refinements.

11 Conclusion

The structural functional can be derived at the effective-theory level, conditional on the chosen admissible ensemble and structural constraints, from four principles: admissibility, structural observability, maximum relative entropy, and large-deviation concentration. This yields

$$\mathrm{d}\mathbb{P}_{\mathrm{sel}}(X) \propto \exp[-\mathcal{F}_{\mathrm{struct}}[X]]\,\mathrm{d}\mu_0(X), \qquad \mathcal{F}_{\mathrm{struct}}[X] = \sum_a \lambda_a r_a[X].$$

In local field systems this becomes an integral over a structural density, and the primitive densities are fixed by equation residuals, Ward/Noether constraint residuals, coarse-grained activity, correlation loss, stability loss, and global integrity defects.

This does not complete a derivation from an ultimate microscopic theory of structural selection. It does, however, replace the statement "the origin of $\mathcal{F}_{\mathrm{struct}}$ is open" with a sharper statement:

> $\mathcal{F}_{\mathrm{struct}}$ is the maximum-entropy rate functional associated with coarse-grained structural constraints on the admissible configuration space. For local field theories, the primitive diagnostics follow from the base action, symmetries, coarse-graining map, and fluctuation operator. The remaining task is to calibrate the reference measure, the conjugate multipliers, and the specialised residuals required by each physical domain.

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